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Centre Number

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Student Number

SCEGGS Darlinghurst

2009

HSC Assessment 2

26th June 2009

Extension 1

Mathematics

Assessment Outcomes: HE4, HE6, HE7

Weighting: 30%

General Instructions

- Time allowed – 70 Minutes
- This paper has **four** questions
- Answer on the paper provided
- Start each question on a **new page**
- Write your Student Number at the top of each page
- Attempt **all** questions and show all necessary working
- Marks may be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used
- A table of standard integrals is provided at the back of this paper

Total marks (46)

- Attempt Questions 1 – 4

Question	Communication	Calculus	Reasoning	Marks
1	/1	/2	/5	/10
2	/4	/6	/2	/12
3	/1	/7	/3	/11
4	/1	/5	/3	/13
TOTAL	/7	/20	/13	/46

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46 marks

Attempt Questions 1 – 4

Answer each question on the paper provided

Write your Student Number at the top of each page

Start each question on a new page

	Marks
Question 1 (10 Marks)	
a) Find: $\int \frac{dx}{\sqrt{9-16x^2}}$	2
b) Consider the function $f(x) = e^x - 3x$	
i) Show a root of $f(x) = 0$ lies between $x = 0$ and $x = 1$	1
ii) Apply the 'halving the interval' method twice to find a smaller interval within which the root lies	2
c) Find the exact value: $\sin\left(2\cos^{-1}\left(-\frac{5}{6}\right)\right)$	2
d) Find the term independent of x in the expansion of $\left(2x^3 - \frac{1}{x}\right)^{12}$	3

Question 2 (12 Marks)

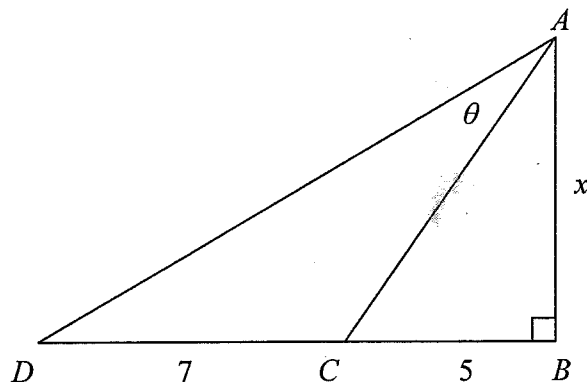
- a) Using the substitution $x = \cos \theta$, evaluate $\int_0^{\frac{1}{2}} 2\sqrt{1-x^2} dx$ 4
- b) i) State the domain and range of the function $y = 2\sin^{-1}\left(\frac{x}{2}-1\right)$ 2
- ii) Sketch the graph of $y = 2\sin^{-1}\left(\frac{x}{2}-1\right)$ clearly labelling all intercepts. 1
- c) i) Use one application of Newton's method to find an approximation of the root of the equation $x^3 - 12x + 2 = 0$ near $x = 3.1$. (to 1 decimal place) 2
- ii) Sketch the curve of $y = x^3 - 12x + 2$ labelling the turning points. 2
- (Do Not Find The x - Intercepts)
- iii) If you choose $x = 1.9$ as your first approximation, explain why you would not arrive at the root near $x = 3.1$ after one application of Newton's Method. 1

Question 3 (11 Marks)

a) Use the substitution $x = u^2$ to find $\int \frac{dx}{\sqrt{x}(1+x)}$ 3

b) Find the greatest coefficient of $(4x^2 + 3x)^9$ 3

- c) In the diagram, DCB is a straight line, $\angle ABC = 90^\circ$, $\angle CAD = \theta$
 $AB = x$, $DC = 7$ and $CB = 5$



i) Show that $\theta = \tan^{-1}\left(\frac{12}{x}\right) - \tan^{-1}\left(\frac{5}{x}\right)$ 1

ii) Show that $\frac{d\theta}{dx} = \frac{5}{x^2 + 25} - \frac{12}{x^2 + 144}$ 2

iii) Find the value of x which maximises θ 2

Question 4 (13 Marks)

- a) Consider the binomial expansion:

3

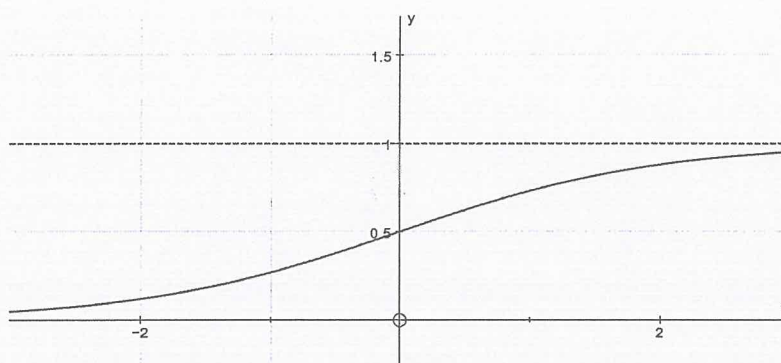
$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

Show that $1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots - (-1)^{n+1} \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}$

- b) Consider the function
- $f(x) = \frac{e^x}{1+e^x}$

- i) Show that
- $y = f(x)$
- is monotonically increasing and hence explain why
- $f^{-1}(x)$
- exists.

2

The graph of $f(x)$ is below

- ii) Sketch
- $y = f^{-1}(x)$

1

- iii) Show algebraically that
- $f^{-1}(x) = \ln\left(\frac{x}{1-x}\right)$

2

- iv) Find
- $f^{-1}\left(\frac{1}{2}\right)$
- and
- $f^{-1}\left(\frac{5}{6}\right)$

2

- v) Evaluate
- $\int_{\frac{1}{2}}^{\frac{5}{6}} f^{-1}(x) dx$

3

End Of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Extension 1 Mathematics Assessment Task 2 2009 - Solutions

Q1 a) $\int \frac{dx}{\sqrt{9-16x^2}} = \frac{1}{4} \int \frac{dx}{\sqrt{9/16-x^2}} \quad \checkmark$
 $= \frac{1}{4} \sin^{-1}\left(\frac{4x}{3}\right) + C \quad \checkmark$
 calc-2

- those who manipulated the integral algebraically did better than those who remembered the formula

b) i) $f(0) = e^0 - 3 \times 0 = 1 > 0$
 $f(1) = e^1 - 3 \times 1 = e - 3 \approx 0.718 < 0$

since $f(0) > 0$ and $f(1) < 0$ and $y=f(x)$ is continuous a root lies between $x=0$ and $x=1$ $\checkmark$
 conn-1

- must mention that $f(x)$ is continuous.

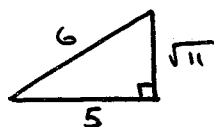
ii)

x	0	1	$\frac{1}{2}$	$\frac{3}{4}$
$f(x)$	1	-0.28	0.49	-0.133

$\checkmark \checkmark$

$\therefore$ root lies between 0.5 and 0.75.

c) let $\alpha = \cos^{-1}\left(-\frac{5}{6}\right) \quad \frac{\pi}{2} < \alpha < \pi$



$\therefore \sin \alpha = \frac{\sqrt{11}}{6} \quad \checkmark$

$\sin 2\alpha = 2 \sin \alpha \cos \alpha$

$= 2 \times \frac{\sqrt{11}}{6} \times -\frac{5}{6}$

$= -\frac{5\sqrt{11}}{18} \quad \checkmark$

Reas-2

- don't well just be careful of silly mistakes.

d) $\left(2x^3 - \frac{1}{x}\right)^{12} = \sum_{r=0}^{12} \binom{12}{r} (2x^3)^{12-r} \left(-\frac{1}{x}\right)^r$

find r : $(2x^3)^{12-r} \left(-\frac{1}{x}\right)^r = x^0$

$36 - 3r - r = 0 \quad \checkmark$

$36 - 4r = 0$

$4r = 36$

$r = 9 \quad \checkmark$

$\therefore$ the term is $\binom{12}{9} (2x^3)^3 \left(-\frac{1}{x}\right)^9$

$= -1760 \quad \checkmark$

Reas-3

DON'T FORGET THE NEGATIVE SIGN.

Q2 a) $\int_0^{\frac{1}{2}} 2\sqrt{1-x^2} dx$ let $x = \cos \theta$ $\frac{1}{2} = \cos \theta$ $0 = \cos \theta$
 $dx = -\sin \theta d\theta$ $\theta = \frac{\pi}{3}$ $\theta = \frac{\pi}{2}$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -2\sin \theta \sqrt{1-\cos^2 \theta} d\theta \quad \checkmark$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 2\sin^2 \theta d\theta$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta \quad \checkmark$$

$$= \left[\theta - \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \quad \checkmark$$

$$= \left(\frac{\pi}{2} - \frac{1}{2} \sin \pi \right) - \left(\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right)$$

$$= \left(\frac{\pi}{2} - 0 \right) - \left(\frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right)$$

$$= \frac{\pi}{6} + \frac{\sqrt{3}}{4} \quad \checkmark$$

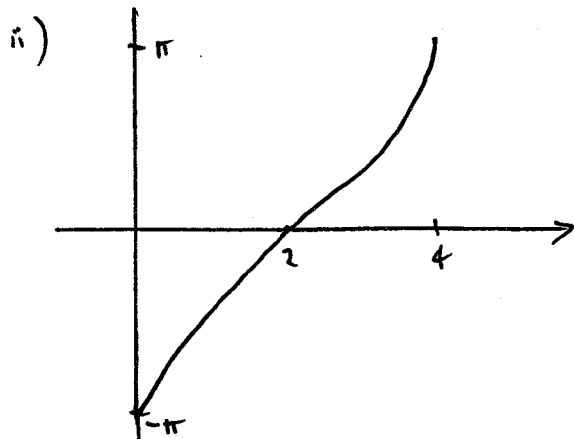
Calc-4

b) i) D: $-1 \leq \frac{x-1}{2} \leq 1$ R: $-\pi \leq y \leq \pi$ $\checkmark$

$$0 \leq \frac{x}{2} \leq 2$$

$$0 \leq x \leq 4 \quad \checkmark$$

Reas-2



$\checkmark$ Conn-1

c) i) let $f(x) = x^3 - 12x + 2$

$$f'(x) = 3x^2 - 12$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

• With the substitution $x = \cos \theta$, when $x=0$ $\theta = \frac{\pi}{2}$ NOT 1!!!

• Many forgot to substitute dx

• The majority of students could not correctly substitute and ended up with a much easier integral - and this unfortunately could not get you any marks. You needed to integrate $\sin^2 \theta$.

Be careful to draw the $\cos^{-1}x$ shape (& not the $\cos x$ shape) $\int$

$$\begin{aligned}
 x_1 &= 3.1 - \frac{[(3.1)^3 - 12(3.1) + 2]}{3(3.1)^2 - 12} \quad \checkmark \\
 &= 3.1 - \frac{(-5.409)}{16.83} \\
 &= 3.4 \quad \checkmark
 \end{aligned}$$

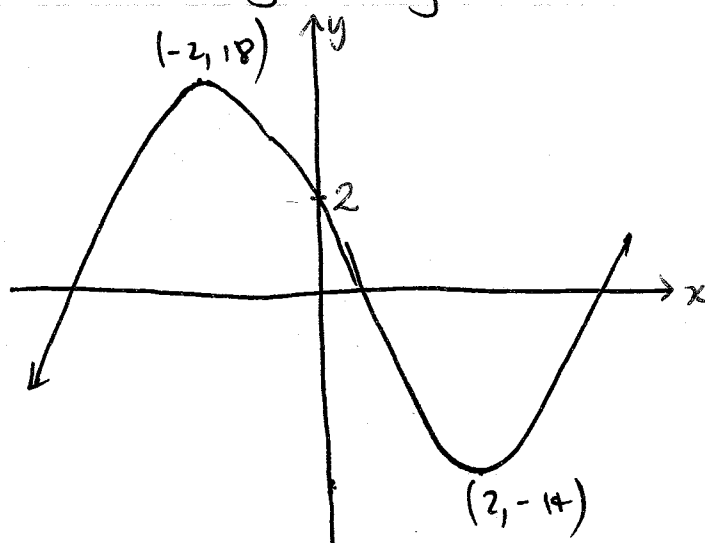
Calc-2

Almost 100% of students got this correct.

ii) turning pts: $\frac{dy}{dx} = 0$

$$\begin{aligned}
 3x^2 - 12 &= 0 \\
 3(x-2)(x+2) &= 0 \\
 x=2 \quad x=-2
 \end{aligned}$$

$$y = -14 \quad y = 18 \quad \checkmark$$



Conn-2

Mostly done well ... but I'm surprised how many did not even calculate the y intercept.

iii) As it is on the other side of the turning point the tangent would get the intercept near another root. $\checkmark$ (Conn-1)

Just because $x=1.9$ is near the turning point doesn't mean anything, the point is it's on the wrong side. Also, the tangent while almost horizontal, will still cut the x axis

Q3 a) $\int \frac{dx}{\sqrt{x}(1+x)}$ $x = u^2$

$$\begin{aligned}
 &= \int \frac{2u du}{u(1+u^2)} \quad \checkmark \\
 &= \int \frac{2 du}{1+u^2} \\
 &= 2 \tan^{-1} u + C \quad \checkmark \\
 &= 2 \tan^{-1} \sqrt{x} + C \quad \checkmark
 \end{aligned}$$

Calc-3

- Students need to recognize integrals. So many didn't see this was an inverse tan solution.

$$b) (4n^3 + 3n)^9 \quad T_{k+1} = {}^9C_k (4n^3)^{9-k} (3n)^k$$

$$C_{k+1} = {}^9C_k (4)^{9-k} (3)^k \quad C_k = {}^9C_{k-1} 4^{9-(k-1)} 3^{k-1}$$

$$\frac{C_{k+1}}{C_k} = \frac{\frac{9!}{(9-k)!k!} \times 4^{9-k} 3^k}{\frac{9!}{[9-(k-1)]!(k-1)!} 4^{9-(k-1)} 3^{k-1}}$$

$$= \frac{10-k}{k} \cdot \frac{3}{4}$$

$$= \frac{30-3k}{4k}$$

$$\therefore \frac{C_{k+1}}{C_k} > 1$$

$$C_k \quad 30-3k > 4k$$

$$7k < 30$$

$$k < 4\frac{2}{7}$$

$$k = 4$$

$$(\text{coeff}) = {}^9C_4 \times 4^5 \times 3^4 = 10450944$$

Recall - 3

$$c) i) \tan \angle CAB = \frac{5}{n} \quad \tan \angle DAB = \frac{12}{n}$$

$$\angle CAB = \tan^{-1}\left(\frac{5}{n}\right) \quad \angle DAB = \tan^{-1}\left(\frac{12}{n}\right)$$

$$\theta = \angle DAB - \angle CAB$$

$$= \tan^{-1}\left(\frac{12}{n}\right) - \tan^{-1}\left(\frac{5}{n}\right) \quad (\text{corr-1})$$

$$ii) \theta = \tan^{-1}\left(\frac{12}{n}\right) - \tan^{-1}\left(\frac{5}{n}\right)$$

$$\frac{d\theta}{dn} = \frac{-12n^{-2}}{1 + \left(\frac{12}{n}\right)^2} - \frac{-5n^{-2}}{1 + \left(\frac{5}{n}\right)^2}$$

$$= \frac{-12/n^2}{1 + \frac{144}{n^2}} + \frac{5/n^2}{1 + \frac{25}{n^2}}$$

$$= \frac{-12/n^2}{\frac{n^2 + 144}{n^2}} + \frac{5/n^2}{\frac{n^2 + 25}{n^2}}$$

$$= \frac{5}{n^2 + 25} - \frac{12}{n^2 + 144}$$

Done very well.
Just reminder to
take your time as
one algebraic mistake
can be costly.

- students thought
this was harder
than it really was
Read the question
carefully

- In a show question
you need to give
a more detailed
solution

iii) find max $\frac{dn}{d\theta} = 0$

$$\frac{5}{x^2+25} - \frac{12}{x^2+144} = 0$$

$$\frac{5}{x^2+25} = \frac{12}{x^2+144}$$

$$5x^2 + 720 = 12x^2 + 300$$

$$7x^2 = 420$$

$$x^2 = 60$$

$$x = \pm\sqrt{60} \quad x = \sqrt{60} \quad \checkmark \text{ as } x \text{ represents length.}$$

test $x = \sqrt{60}$

x	7	$\sqrt{60}$	8
$\frac{dn}{dx}$	0.0053	0	-0.0015

$\therefore x = \sqrt{60}$ gives a maximum. Calc-2 $\checkmark$

Q4 a) $\int (1+x)^n = \int 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$

$$\frac{(1+x)^{n+1}}{n+1} + C = x + \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} + \dots + \binom{n}{n}\frac{x^{n+1}}{n+1} \quad \checkmark$$

let $x=0$ to find C

$$C = -\frac{1}{n+1}$$

$$\therefore \frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1} = x + \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} + \dots + \binom{n}{n}\frac{x^{n+1}}{n+1} \quad \checkmark$$

let $x = -1$

$$\frac{-1}{n+1} = -1 + \frac{1}{2}\binom{n}{1} - \frac{1}{3}\binom{n}{2} + \dots + \frac{(-1)^{n+1}}{n+1}\binom{n}{n} \quad \checkmark$$

$$1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots - \frac{(-1)^{n+1}}{n+1}\binom{n}{n} = \frac{1}{n+1}$$

Reas-3

This part was done well if you got up to it.

This question was very well done.

b) i) $y = \frac{e^x}{e^x + 1}$

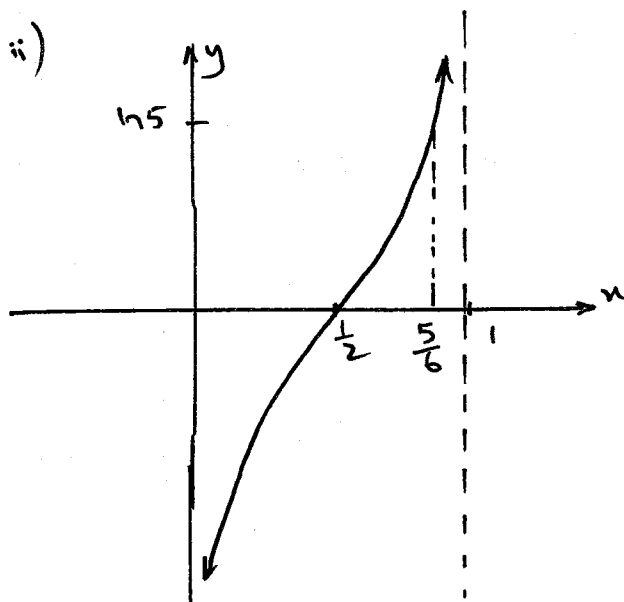
$$\begin{aligned} \frac{dy}{dx} &= \frac{(e^x + 1)e^x - e^x \cdot e^x}{(e^x + 1)^2} \\ &= \frac{e^{2x} + e^x - e^{2x}}{(e^x + 1)^2} \\ &= \frac{e^x}{(e^x + 1)^2} \end{aligned}$$

since $e^x > 0$ for all x

$$\frac{dy}{dx} > 0 \text{ for all } x$$

$\therefore$ the curve of $y = f(x)$ is monotonically increasing and hence has an inverse function as it satisfies the horizontal line test.

Calc-2



iii) $y = \frac{e^x}{1 + e^x}$

interchange x and y

$$x = \frac{e^y}{1 + e^y}$$

$$x + x e^y = e^y$$

$$\begin{aligned} x &= e^y - x e^y \\ &= e^y(1 - x) \end{aligned}$$

Note: every x has a unique y is NOT the horizontal line test (it's the vertical line test)

There were some very sloppy graphs Use a pencil & a ruler, label the x intercept and label the asymptote

There was a lot of fudging for that 2nd mark!

$$e^y = \frac{x}{1-x}$$

$$y = \ln\left(\frac{x}{1-x}\right)$$

$$iv) f^{-1}\left(\frac{1}{2}\right) = \ln\left(\frac{\frac{1}{2}}{1-\frac{1}{2}}\right)$$

$$= \ln 1$$

$$= 0 \quad \checkmark$$

$$f^{-1}\left(\frac{5}{6}\right) = \ln\left(\frac{\frac{5}{6}}{1-\frac{5}{6}}\right)$$

$$= \ln\left(\frac{\frac{5}{6}}{\frac{1}{6}}\right) \quad \checkmark$$

$$= \ln 5$$

$$v) \int_{\frac{1}{2}}^{\frac{5}{6}} f^{-1}(x) dx = \underbrace{\frac{5}{6} \ln 5}_{\text{Area } \square} - \int_0^{\frac{5}{6}} \frac{e^x}{1+e^x} dx \quad \checkmark$$

$$= \frac{5}{6} \ln 5 - [\ln(1+e^x)]_0^{\frac{5}{6}} \quad \checkmark$$

$$= \frac{5}{6} \ln 5 - [\ln(1+e^{\frac{5}{6}}) - \ln(1+e^0)]$$

$$= \frac{5}{6} \ln 5 - (\ln 6 - \ln 2)$$

$$= \frac{5}{6} \ln 5 - \ln 3. \quad \checkmark$$

Calc-3

There were 2 easy marks picked up by most in a rush at the end. Well done.

lots of careless mistakes in the last substitution which was a pity.